

Customer Churn Analysis Dashboard using Power BI

Executive Summary

Customer retention is one of the most critical factors influencing the long-term success of a telecommunications company. Acquiring new customers is often more expensive than retaining existing ones. This project aims to analyze customer churn patterns and identify the key factors driving customer attrition using Power BI.

The dashboard provides actionable insights into customer demographics, service usage, contract preferences, revenue contribution, and churn behavior, enabling data-driven decision-making to improve customer retention strategies.

Business Objective

The primary objective of this project is to:

- Analyze customer churn behavior.
 - Identify high-risk customer segments.
 - Understand the impact of contract types and services on churn.
 - Evaluate revenue patterns across customer groups.
 - Provide recommendations to reduce customer attrition and improve customer loyalty.
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Dataset Overview

The dataset contains customer information including:

- Customer Demographics
- Contract Details
- Internet Services
- Payment Methods
- Monthly Charges
- Total Charges
- Customer Tenure
- Churn Status

The dataset was cleaned, transformed, and modeled in Power BI to create an interactive business intelligence dashboard.

Dashboard Structure

1. KPI Overview

The KPI section provides a high-level summary of customer performance.

Key Metrics:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Average Monthly Charges
- Average Tenure

This section helps management quickly assess overall business health and customer retention performance.

2. Customer Profile Analysis

This section explores customer demographics and their relationship with churn.

Analysis Included:

- Gender Distribution
- Senior Citizen Analysis
- Partner Status
- Dependents Analysis

Key Finding:

Senior citizens and customers without dependents showed a relatively higher tendency to churn compared to other customer segments.

3. Contract & Service Analysis

This section investigates how service subscriptions and contract types influence customer retention.

Analysis Included:

- Contract Type vs Churn
- Internet Service vs Churn
- Payment Method Analysis
- Online Security
- Tech Support
- Streaming Services

Key Finding:

Customers on Month-to-Month contracts exhibited significantly higher churn rates compared to customers with One-Year and Two-Year contracts.

4. Revenue & Charges Analysis

This section focuses on revenue generation and customer value.

Analysis Included:

- Revenue by Contract Type
- Revenue by Internet Service
- Monthly Charges Distribution
- Total Charges Analysis

Key Finding:

Customers with higher monthly charges were more likely to churn, indicating potential dissatisfaction with perceived value.

5. Churn Risk & Tenure Analysis

This section identifies customer groups most vulnerable to churn.

Analysis Included:

- Tenure Group Analysis
- Churn Rate by Tenure
- Monthly Charges vs Tenure

Key Finding:

Customers within their first year of service demonstrated the highest churn risk, highlighting the importance of early customer engagement and retention programs.

Key Business Insights

1. Month-to-Month contracts contribute the highest percentage of customer churn.
 2. Customers with shorter tenure are significantly more likely to leave the company.
 3. Fiber Optic service users exhibit higher churn rates compared to other service categories.
 4. Customers lacking Online Security and Tech Support services show increased churn behavior.
 5. Higher monthly charges correlate with increased customer attrition.
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Recommendations

Based on the analysis, the following recommendations are proposed:

- Encourage customers to switch from Month-to-Month contracts to long-term plans.
 - Improve onboarding experiences for new customers.
 - Offer loyalty programs targeting customers within their first year.
 - Bundle Online Security and Tech Support services to improve retention.
 - Review pricing strategies for high-cost customer segments.
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Conclusion

This Power BI dashboard successfully transformed raw customer data into actionable business insights. The analysis revealed critical churn drivers related to contract types, service subscriptions, tenure, and pricing.

By leveraging these insights, businesses can implement targeted retention strategies, improve customer satisfaction, and enhance long-term profitability.

Tools Used

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel / CSV Dataset

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